Analysis:

$$T(n) = \sum_{i=1}^{n} \sum_{j=i}^{n} \sum_{k=i}^{j-1} 1.$$

$$= \sum_{i=1}^{n} \sum_{j=i}^{n} (j-1) - i + 1$$

$$T(n) = \sum_{i=1}^{n} \sum_{j=i}^{n} (j-i)$$

$$T(n) = \sum_{i=1}^{n} \left(0+1+2+3+\cdots+(n-i)\right)$$

$$= \sum_{i=1}^{n} \frac{(n-i)(n-i+1)}{2}$$

$$= \frac{1}{2}\sum_{i=1}^{n}(n-i)^2 + \frac{1}{2}\sum_{i=1}^{n}(n-i)$$

$\frac{n(n+1)(2n+1)}{6}$ ⟷ vse

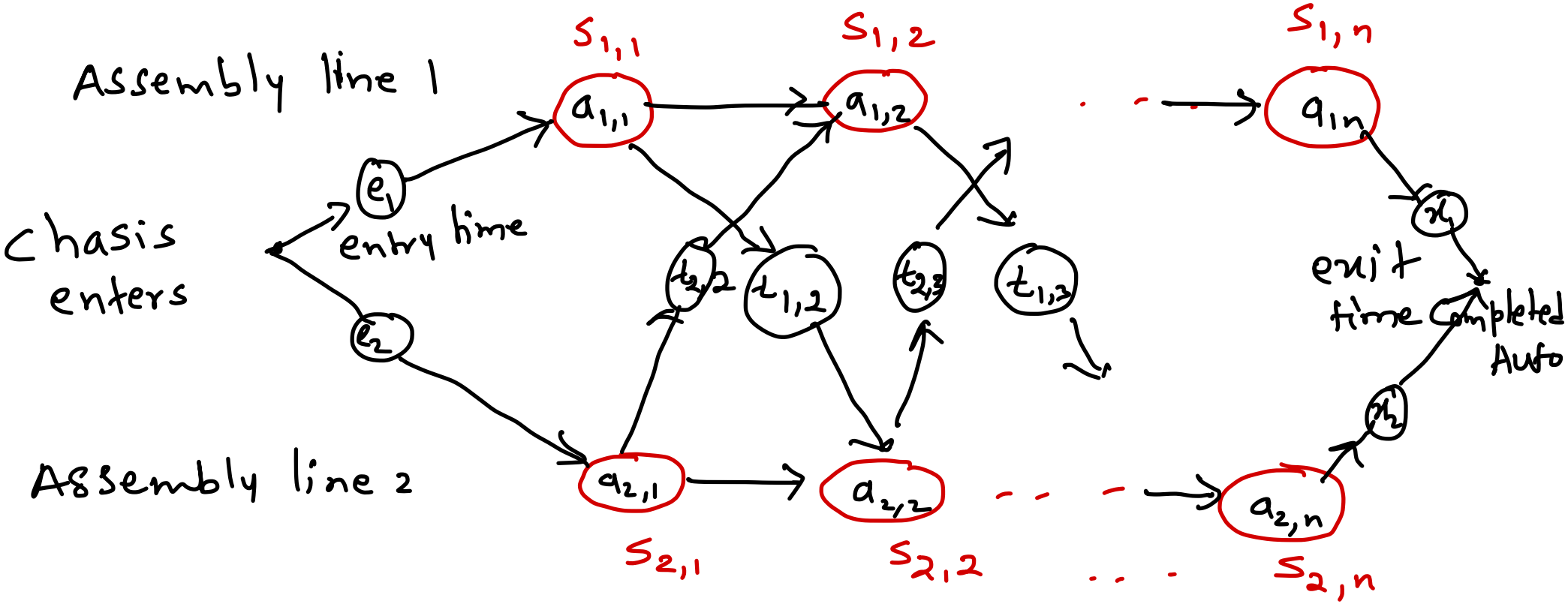
$$= \frac{1}{2}\left[(n-1)^2 + (n-2)^2 + \cdots + 0^2\right] + \frac{1}{2}\left[(n-1) + (n-2) + \cdots + 0\right]$$

$$= \frac{1}{2} \cdot \frac{(n-1)(n)(2n-1)}{6} + \frac{1}{2} \cdot \frac{1}{2}(n-1)n$$

$$= \frac{n^3 - n}{6}$$

$$\in \Theta(n^3).$$

Assembly-line scheduling:



Question:

$f_i[j]$ = fastest time to get a chasis from starting point

through station S_{ij}

$$f^* = \min \{ f_1[n] + x_4 , f_2[n] + x_2 \}$$

$$= ?$$

Brute-Force: $\{1, 2, \ldots, n\}$ $\quad \min \quad 2^n$

$\{1, 2, \ldots n\}$

$$f_1[j] = \begin{cases} e_1 + a_{1,1} & \text{if } j = 1. \\ \min\{ f_1[j-1] + a_{1,j}, & \text{if } j \geq 2 \\ \quad f_2[j-1] + a_{1,j} + t_{2,j} \} \end{cases}$$

$$f_2[j] = \begin{cases} e_2 + a_{2,1} & \text{if } j = 1 \\ \min\{ f_2[j-1] + a_{2,j}, & \text{if } j \geq 2. \\ \quad f_1[j-1] + a_{2,j} + t_{1,j} \} \end{cases}$$

$r_i(j)$ = no. of references made to $f_i[j]$ $i = 1, 2.$

$$r_1(j) = r_1(j-1) + r_2(j-1)$$
$$r_2(j) = r_2(j-1) + r_1(j-1)$$

} Solve (I)

Good guess: $r_1(j-1) = r_2(j-1)$.

$$r_1(j) = r_1(j-1) + r_1(j-1)$$
$$= 2\, r_1(j-1)$$
$$= 2\left(2 \cdot r_1(j-2)\right)$$
$$= 2^2\, r_1(j-2)$$
$$= 2^2\left(2 \cdot r_1(j-3)\right)$$
$$= 2^3\, r_1(j-3)$$

$$= 2^{j-1} r_1(j-(j-1))$$

$$= 2^{j-1} r_1(1)$$

$$= 2^{j-1}.$$

Claim: $r_1(j) = 2^{j-1}$ is the solution of (I).

$$r_2(j) = 2^{j-1}$$

Substitute in (I)

$$r_1(j) = r_1(j-1) + r_2(j-1)$$

$$= 2^{j-2} + 2^{j-2}$$

$$= 2\left(2^{j-2}\right)$$

$$= 2^{j-1}.$$

$$r_2(j) = r_2(j-1) + r_1(j-1)$$

$$= 2^{j-2} + 2^{j-2}$$

$$= 2^{j-1}$$ //

(or use induction to prove the claim).